

Ch. 1 — P02 (D) Design a Simulation Study

Problem Bank | Chapter 1 | Type: D | Difficulty:

Chapter: 1 Type: D Difficulty:

A regional airport is considering adding a second security checkpoint lane to reduce passenger wait times during peak hours.

Task: Design a simulation study (do *not* build the model — describe the study).

Your design document must address:

1. **Objective statement:** What specific question will the simulation answer? State it precisely.
2. **Scope and boundaries:** What is inside and outside the model?
3. **Inputs:** What data would you need to collect, and from where?
4. **Outputs / performance measures:** What metrics will you report to the airport manager?
5. **Scenarios:** What system configurations will you compare?
6. **Assumptions:** List at least four simplifying assumptions and justify each.
7. **Validation plan:** How would you validate the model before using it for decisions?
8. **Communication plan:** Who is the audience? What format will the final report take?

Hint: Follow the 12-step simulation study process from Chapter 1.